

Customer Churn Prediction & Retention Analytics

for Service-Oriented Businesses

*A Business-Oriented AI & Analytics System using Machine Learning,
Customer Segmentation and Explainable AI*

Project Domain	Business Analytics, Artificial Intelligence & Machine Learning
Project Type	Applied AI / Business Analytics (Data Product)
Techniques Used	Logistic Regression, Random Forest, XGBoost, SVM, K-Means Segmentation, SHAP-based Explainable AI
Deployment	Streamlit interactive analytics dashboard
Dataset	Customer Churn Prediction dataset (Kaggle, public)
Primary Deliverable	Churn risk scoring engine + retention decision-support dashboard
Report Version	Update 1 (revised scope — business-oriented AI/analytics project)
Date	10 August 2026

Table of Contents

1. Abstract
 2. Introduction
 3. Problem Statement
 4. Aim
 5. Objectives
 6. Motivation
 7. Scope of the Project
 8. Literature Survey
 9. Existing Methods and Their Drawbacks
 10. Research Gaps Identified
 11. Proposed System
 12. System Architecture
 13. Detailed Methodology
 14. Dataset Description
 15. Hardware and Software Requirements
 16. Applications
 17. Expected Output
 18. Advantages
 19. Work Breakdown Structure and Timeline
 20. Conclusion
- References

Revision note (Update 1): This version upgrades the project from a purely predictive machine-learning exercise into an end-to-end, business-oriented AI and analytics solution. Newly added components are customer segmentation, engineered behavioural features, Logistic Regression as an interpretable benchmark, per-customer churn probability scoring, Explainable AI with feature importance and SHAP reason codes, a high-risk customer dashboard, prescriptive business recommendations, and a quantified revenue / retention impact model.

1. Abstract

Customer churn — the loss of existing customers to competitors or to service discontinuation — is one of the most financially damaging problems faced by service-oriented businesses such as telecommunications operators, banks, insurers, subscription platforms and SaaS providers. Because acquiring a new customer typically costs several times more than retaining an existing one, even a marginal reduction in the churn rate translates directly into measurable profit. Most academic churn projects, however, stop at reporting a classification accuracy figure and never answer the question a business actually asks: which specific customers are about to leave, why are they leaving, what should we do about it, and how much money will that save?

This project proposes a complete, business-oriented AI and analytics system for customer churn prediction and retention management. The system begins with rigorous data cleaning and exploratory data analysis of a public customer churn dataset, followed by domain-driven feature engineering (tenure bands, average revenue per user, charge ratios, service-adoption counts) and unsupervised customer segmentation using K-Means clustering over RFM-style value indicators. Four supervised classifiers — Logistic Regression, Random Forest, XGBoost and a Support Vector Machine baseline — are trained and benchmarked using accuracy, precision, recall, F1-score, ROC-AUC and PR-AUC, with class imbalance addressed through stratified sampling and class weighting. The best model is used to produce a calibrated churn probability for every customer rather than a simple yes/no label, allowing customers to be ranked and banded into low, medium and high risk tiers.

Model transparency is provided through an Explainable AI layer that combines global feature-importance analysis with SHAP (SHapley Additive exPlanations) values, generating human-readable reason codes for each individual prediction — for example, "month-to-month contract", "short tenure" or "high monthly charges relative to segment". These explanations drive a prescriptive recommendation engine that maps each risk driver to a concrete retention action. Finally, a business impact module combines churn probability with customer value to compute revenue at risk, expected saved revenue and retention-campaign ROI. The entire pipeline is deployed as an interactive Streamlit dashboard that lists high-risk customers, visualises segment-level churn behaviour and exports prioritised target lists for the retention team. The system targets a classification accuracy above 85% and a ROC-AUC above 0.85, while delivering decision-ready, financially quantified output.

Keywords

Customer churn prediction; retention analytics; business intelligence; machine learning; Logistic Regression; Random Forest; XGBoost; customer segmentation; K-Means clustering; Explainable AI; SHAP; churn probability; revenue at risk; customer lifetime value; Streamlit dashboard.

2. Introduction

In a service-driven economy, revenue is recurring rather than transactional. A telecom subscriber, a bank account holder or a SaaS subscriber generates income month after month, so the value of that relationship is realised over its lifetime rather than at the point of sale. Consequently, the single most important determinant of long-term profitability for such organisations is not how many customers they acquire but how many they keep. Customer churn — also called customer attrition — refers to the event in which a customer terminates their relationship with a service provider, either by cancelling a subscription, switching to a competitor, or simply becoming inactive.

The financial asymmetry between acquisition and retention is well established in marketing literature: attracting a new customer generally costs between five and twenty-five times more than retaining an existing one, and existing customers spend more, adopt more services and refer more prospects. In mature, saturated markets such as Indian telecom or retail banking, where the pool of new customers is limited, retention is effectively the only sustainable growth lever. A churn rate reduction of even two or three percentage points across a large subscriber base can represent crores of rupees in preserved annual revenue.

Historically, churn management has been reactive. Businesses discovered churn only after it happened, through monthly attrition reports, and responded with broad, untargeted win-back campaigns or blanket discounts. Where analytics was used at all, it took the form of descriptive statistics, static rule-based flags (for example, "flag any customer who called support more than three times") or simple regression on a handful of variables. Such approaches struggle with the reality of modern customer data: hundreds of correlated attributes, strongly non-linear interactions between them, and heavily imbalanced outcome classes in which churners form a minority of the population.

Machine learning changes this picture fundamentally. Supervised classification algorithms can learn complex, non-linear behavioural signatures from historical customer records and estimate, for each individual customer, the probability that they will churn in the next billing cycle. This converts churn management from a reactive, post-mortem activity into a proactive, predictive one. Ensemble methods such as Random Forest and gradient-boosted trees such as XGBoost have repeatedly been shown in recent literature to deliver state-of-the-art performance on tabular customer data, while interpretable models such as Logistic Regression remain valuable as transparent, well-calibrated baselines.

However, predictive accuracy alone does not create business value. A model that reports 87% accuracy but cannot explain its reasoning, cannot be inspected by a retention manager, and cannot be tied to a rupee figure will not be adopted operationally. Recent research therefore emphasises Explainable AI — particularly SHAP-based attribution — to expose the drivers behind each individual prediction, and profit-driven evaluation that scores models on the money they save rather than on abstract metrics.

This project responds directly to that shift. It is designed not as a basic machine-learning exercise but as a business-oriented AI and analytics product. It integrates data cleaning and exploratory analysis, customer segmentation, feature engineering, multi-model benchmarking, per-customer churn probability prediction, explainable feature attribution, a high-risk customer dashboard, prescriptive retention recommendations and an estimated revenue and retention impact model into a single, deployable decision support system.

3. Problem Statement

Service-oriented organisations hold vast quantities of customer data — demographics, contract terms, service subscriptions, billing history and support interactions — yet remain largely unable to convert that data into timely, targeted retention action. The core problem this project addresses can be stated as follows:

“Given historical customer records containing demographic, contractual, service-usage and billing attributes, how can an organisation accurately identify, in advance, which individual customers are at high risk of churning; understand the specific reasons driving each customer's risk; prioritise them by their financial value to the business; and quantify the revenue that a targeted retention intervention would preserve?”

3.1 Specific Challenges

- **Scale and complexity of data** — Customer datasets contain a large number of mixed categorical and numerical attributes whose interactions are non-linear and not discoverable by manual inspection.
- **Class imbalance** — Churners typically constitute only 20–30% of the customer base, causing naive models to achieve deceptively high accuracy while failing to detect the minority class that actually matters.
- **Binary output is not actionable** — A yes/no churn label gives the retention team no way to prioritise. Two customers labelled 'will churn' may have a 51% and a 96% probability and vastly different values.
- **Opacity of high-accuracy models** — Ensemble and boosting models behave as black boxes. Without explanation, business users cannot trust the output or design a targeted response.
- **No link to money** — Conventional projects report accuracy and ROC-AUC but never translate predictions into revenue at risk, saved revenue or campaign ROI, which is the language management actually uses.
- **Absence of a usable interface** — Models that live inside notebooks are never adopted. Retention teams need a dashboard listing at-risk customers with the reasons and the recommended action.
- **One-size-fits-all retention** — Without customer segmentation, the same offer is pushed to all at-risk customers, wasting budget on low-value customers and under-serving high-value ones.

Therefore, there exists a clear need for an intelligent, explainable and financially grounded churn prediction and retention analytics system that closes the gap between statistical prediction and business decision-making.

4. Aim

The aim of this project is to design, develop and deploy an end-to-end, business-oriented AI and analytics system that predicts the probability of customer churn in service-oriented businesses, explains the drivers behind every prediction, segments and prioritises at-risk customers by their financial value, recommends targeted retention actions, and quantifies the expected revenue and retention impact of those actions through an interactive web-based decision-support dashboard.

5. Objectives

5.1 Primary Objectives

1. To acquire, clean and profile a real-world customer churn dataset, handling missing values, inconsistent data types, duplicates and outliers, and to perform structured exploratory data analysis (EDA) that establishes the baseline churn rate and the key behavioural patterns associated with churn.
2. **To engineer meaningful business-oriented features** — including tenure bands, average revenue per user (ARPU), total-to-monthly charge ratio, number of subscribed services and contract-risk indicators — that improve model performance and interpretability beyond the raw dataset columns.
3. To perform unsupervised customer segmentation using K-Means clustering over RFM-style value indicators, and to profile the churn behaviour, revenue contribution and risk exposure of each resulting segment.
4. **To implement, train and tune four supervised classification models** — Logistic Regression, Random Forest, XGBoost and a Support Vector Machine baseline — for churn classification.
5. To benchmark and compare these models on accuracy, precision, recall, F1-score, ROC-AUC and PR-AUC, and to select the best-performing model, targeting accuracy above 85%.
6. To generate a calibrated churn probability for every individual customer and to band customers into low, medium and high risk tiers rather than producing a simple binary label.
7. To build an Explainable AI layer using global feature importance and SHAP values that produces human-readable reason codes explaining why each individual customer is flagged as at-risk.
8. To develop an interactive Streamlit dashboard that displays the ranked list of high-risk customers, segment-level KPIs, model comparison charts and drill-down explanations, with CSV export.
9. To generate prescriptive business recommendations that map each identified churn driver to a concrete, cost-aware retention action.

10. To build a business impact model that estimates revenue at risk, expected saved revenue, retention-campaign ROI and the projected reduction in the overall churn rate.

5.2 Analytical Variables

Variable Type	Variables	Role in the Study
Independent Variables (IV)	Tenure, Monthly Charges, Total Charges, Contract Type, Payment Method, Internet Service Type, Tech Support, Online Security, Paperless Billing, Senior Citizen, Partner/Dependents	Predictors used to model churn behaviour
Engineered Variables	Tenure band, ARPU, Total/Monthly charge ratio, Number of subscribed services, Contract risk flag, Customer value tier	Derived features that add business meaning and lift model performance
Segmentation Variables	Tenure (recency proxy), Number of services (frequency proxy), Monthly Charges (monetary proxy)	Inputs to K-Means clustering for customer segmentation
Dependent Variable (DV)	Churn status (Yes = 1 / No = 0) and the associated churn probability $p(\text{churn}) \in [0, 1]$	Target variable predicted by the models
Business Output Variables	Risk band, Revenue at Risk (RaR), Expected saved revenue, Retention ROI	Derived business metrics used for decision-making

6. Motivation

The motivation for this project is both economic and technical.

6.1 Economic Motivation

- **Retention is cheaper than acquisition** — Industry studies consistently place the cost of acquiring a new customer at five to twenty-five times the cost of retaining an existing one; retained revenue therefore carries a far higher margin.
- **Small churn reductions compound** — On a base of 7,000 customers with an average monthly charge of approximately ₹5,400 (US\$64.76 equivalent), reducing churn from 26.5% to 22% preserves roughly 315 customers per cycle, worth over ₹2 crore in annualised recurring revenue.
- **Revenue is concentrated** — A minority of high-value customers generates a disproportionate share of revenue, so churn among that minority is far more damaging than churn among low-value customers. Value-weighted prioritisation is therefore essential.
- **Retention budgets are finite** — Retention offers cost money. Targeting the wrong customers (those who would have stayed anyway, or those not worth saving) destroys campaign ROI.

6.2 Technical and Academic Motivation

- **Machine learning captures non-linear behaviour** — Ensemble and boosting methods discover interaction effects (for example, short tenure combined with month-to-month contract and electronic-cheque payment) that no manual rule set would encode.
- **Explainable AI has matured** — SHAP and related attribution methods now make it practical to open the black box and give each prediction a defensible, per-customer explanation, which is a precondition for operational and regulatory acceptance.
- **Analytics must become prescriptive** — The field is moving from descriptive (what happened) through predictive (what will happen) to prescriptive analytics (what should we do about it). This project is deliberately positioned at the prescriptive end.
- **Deployment is the differentiator** — A model in a notebook has no business value. Packaging the pipeline as a live dashboard converts an experiment into a usable data product.

6.3 Personal / Project Motivation

A conventional churn project trains a classifier and reports an accuracy figure. This project is deliberately built as a business analytics and AI product: it answers who will churn, why they will churn, which of them are worth saving, what should be offered to each of them, and how much money the intervention is expected to preserve. That end-to-end orientation is what makes the work valuable to a recruiter, to an evaluator and, most importantly, to a real business.

7. Scope of the Project

7.1 In Scope

- **Domain coverage** — Telecommunications, banking and financial services, insurance, SaaS and subscription platforms, OTT/streaming media, e-commerce loyalty programmes, and gym/utility subscription businesses.
- **Data type** — Structured, tabular customer data containing demographic, contractual, service-usage and billing attributes, supplied as CSV or exported from a CRM/billing system.
- **Analytical scope** — Data cleaning, exploratory data analysis, feature engineering, K-Means customer segmentation, supervised churn classification, model benchmarking, probability calibration, SHAP-based explainability, and business-impact quantification.
- **Modelling scope** — Logistic Regression, Random Forest, XGBoost and SVM, with stratified train/test splitting, cross-validation and class-imbalance handling.
- **Deployment scope** — A single-user/small-team Streamlit web application supporting dataset upload, batch scoring of an entire customer file, single-customer what-if prediction, dashboard visualisation and CSV export of the prioritised target list.
- **Business scope** — Risk banding, retention recommendation generation, revenue-at-risk calculation and retention-campaign ROI estimation.

7.2 Out of Scope (Current Phase)

- Real-time streaming ingestion from live production databases (the system operates on batch CSV data).
- Causal uplift modelling requiring randomised control-group experimental data.
- Automatic execution of retention campaigns (the system recommends; the business acts).
- Deep-learning sequence models (LSTM/GRU) over longitudinal transaction histories.
- Multi-tenant, enterprise-grade authentication, role-based access control and audit logging.

7.3 Future Extensibility

- Integration with live CRM and billing systems through scheduled or API-based data pulls.
- Automated model retraining and drift monitoring as customer behaviour changes over time.
- Uplift / incremental-response modelling to isolate genuinely persuadable customers.
- Full customer lifetime value (CLV) modelling in place of the current ARPU-based value proxy.
- Extension to next-best-offer recommendation and cross-sell / up-sell analytics.

8. Literature Survey

A survey of recent peer-reviewed literature (2022–2026) was conducted across four themes directly relevant to this project: (i) supervised and ensemble machine learning for churn classification, (ii) class-imbalance handling, (iii) Explainable AI applied to churn, and (iv) customer segmentation and value-based retention analytics. The survey confirms three clear trends: gradient-boosted tree ensembles dominate tabular churn prediction; SHAP-based explainability has become standard rather than optional; and the research frontier has moved from raw predictive accuracy towards profit-aware, segment-aware and action-oriented churn management.

8.1 Summary of Reviewed Literature

#	Author(s) / Source & Year	Title / Focus	Method Used	Key Findings	Limitations
1	Habib, Aslam, Ghafoor & Shahzad, SSRN (2025)	Enhancing telecom churn prediction using hybrid ensemble learning and class-imbalance handling	Hybrid ensemble of multiple classifiers with SMOTE-ENN resampling and novel feature engineering	Hybrid ensembles with imbalance correction outperform single classifiers; engineered features add significant lift	Computationally heavy; no explainability layer or business-impact quantification
2	arXiv:2408.16284 (2024)	Adaptive ensemble learning approach for churn prediction in telecommunications	Stacking ensemble of XGBoost, LightGBM, LSTM, MLP and SVM with meta-feature generation	Adaptive stacking yields substantial accuracy gains over individual state-of-the-art models	Very high model complexity; results risk over-optimism; not deployable as a business tool
3	Machine Learning with Applications, Elsevier (2024)	Explaining churn prediction in the telecom industry using tabular ML models	Tabular ML models combined with post-hoc interpretability analysis	Interpretability techniques successfully expose the dominant churn drivers in tabular telecom data	Explanation stops at model level; no prescriptive retention action derived
4	Information (MDPI), 16(7):537 (2025)	Comprehensive evaluation of machine learning and deep learning models for churn prediction	Systematic benchmark of classical ML against deep learning architectures	Well-tuned gradient boosting matches or beats deep learning on tabular churn data at far lower cost	Benchmark-only study; no deployment, segmentation or financial evaluation
5	Data Science in Finance & Economics, AIMS Press, 5(1) (2025)	Prediction of bank credit customer churn with interpretability analysis	XGBoost classifier interpreted using SHAP values	XGBoost reached ~97% across accuracy, precision, recall, F1 and AUC; SHAP identified transaction count, transaction amount and product holdings as dominant drivers	Banking-specific; no customer segmentation or ROI model
6	Scientific Reports / PMC (2026)	Explainable AI-driven churn prediction: multi-model ensemble with SHAP-based feature analysis	Seven ML models incl. XGBoost, Random Forest and MLP with SMOTE and SHAP analysis	XGBoost achieved the best discrimination (AUC-ROC ≈ 0.932 , F1 ≈ 0.84); SHAP identified contract type, tenure and technical support as the top churn contributors	Explanations are not converted into targeted retention strategies or revenue estimates
7	Eng., Technology & Applied Science Research, ETASR (2024)	Churn prediction for telecommunication companies using ML and ensemble methods	Comparative study of classical classifiers and ensemble methods	Ensemble methods consistently outperform single classifiers on telecom churn datasets	No explainability, segmentation, dashboard or business linkage

8	Expert Systems with Applications, Elsevier (2025)	Enhancing retention in online retail through churn prediction: hybrid RFM, K-Means and deep neural network approach	RFM feature construction + K-Means segmentation + LSTM/GRU deep models	Segmenting customers before modelling improves both accuracy and the interpretability of retention strategy; six behavioural segments identified	Retail/transactional data only; deep models are opaque and hard to deploy at small scale
9	Proc. 10th Int. Conf. on Intelligent Information Technology, ACM (2025)	Customer segmentation: automatic K-optimisation and RFM-based K-Means clustering	RFM scoring with automated selection of the optimal number of clusters	Automated K-optimisation produces stable, high-precision, business-interpretable segments	Segmentation only; no predictive churn model attached
10	Scientific Reports (2026)	Hybrid deep learning framework for churn prediction using RFM and embedding clustering	RFM feature engineering + Deep Embedded Clustering + deep classifiers, jointly optimised	Joint segmentation-and-prediction outperforms treating the two tasks separately	Requires large data volumes and heavy computation; limited transparency
11	J. Theoretical & Applied Electronic Commerce Research, MDPI (2026)	Customer segmentation using an extended RFM model and clustering algorithms in e-commerce	Extended RFM (adding interval and standard-deviation attributes) with multiple clustering algorithms	Recency, frequency and purchase-interval variability are strong indicators of churn propensity and loyalty	Descriptive segmentation; no supervised churn classifier or ROI model
12	Ingénierie des Systèmes d'Information, IIETA (2025)	Explainable customer churn prediction in telecom	Tree-ensemble classifiers combined with SHAP-based global and local explanation	Local explanations make individual churn predictions auditable and defensible to business users	Does not close the loop into recommended actions or measurable financial outcome
13	ResearchGate preprint (2025)	Explainable AI approach for telecom churn prediction and retention strategy design using XGBoost and SHAP	XGBoost with global and local SHAP explainability feeding retention strategy design	Demonstrates an actionable bridge from algorithmic output to business decision-making	Retention strategy is qualitative; expected revenue impact is not quantified
14	Wei, Theoretical and Natural Science, vol. 134 (2025)	Comparative analysis of machine learning models for telecom customer churn prediction	Head-to-head comparison of logistic, tree-based and ensemble classifiers	Confirms tree ensembles as the strongest performers while logistic regression remains the most interpretable benchmark	Purely comparative; no deployment or business-value analysis
15	IJERT (2024)	Integration of churn rate prediction and customer lifetime value estimation using temporal convolution networks	Temporal convolution networks jointly modelling churn risk and CLV	Combining churn probability with customer value produces far better retention prioritisation than risk ranking alone	Requires longitudinal time-series data unavailable in standard cross-sectional churn datasets

8.2 Inferences Drawn from the Literature

- **Gradient-boosted ensembles are the benchmark to beat** — XGBoost and related boosting methods repeatedly deliver the highest ROC-AUC on tabular churn data, typically in the 0.84–0.93 range on public datasets, which validates the model selection made in this project.

- **Logistic Regression remains essential** — Almost every comparative study retains it as an interpretable, well-calibrated baseline against which ensemble gains are measured; it is included in this project for the same reason.
- **Class imbalance must be handled explicitly** — Studies that ignore it report inflated accuracy with poor recall on the churn class; stratification, class weighting or SMOTE-family resampling is standard practice.
- **Contract type, tenure and support-related attributes dominate** — Across banking, telecom and retail studies, SHAP consistently ranks contract/commitment length, tenure and service-support variables among the strongest churn drivers.
- **Explainability is now expected, not optional** — Post-2024 literature almost universally includes SHAP or an equivalent attribution method.
- **Segmentation before prediction improves outcomes** — RFM-style segmentation combined with clustering yields both better models and more targeted retention strategies.
- **The unresolved gap is action and money** — Very few studies close the loop from prediction to a specific recommended intervention with an estimated financial return. That gap is the core contribution of this project.

9. Existing Methods and Their Drawbacks

9.1 Existing Methods in Practice

Category	Method	Description
Manual / Descriptive	Periodic churn reporting	Monthly or quarterly attrition dashboards showing churn that has already occurred, with no forward-looking capability.
Manual / Descriptive	Manual customer segmentation	Analysts group customers by intuition or single attributes such as plan type or region.
Rule-Based	Static business rules	Hard-coded thresholds such as 'flag customers with more than three support calls' or 'flag month-to-month customers with tenure below six months'.
Rule-Based	Exit surveys / feedback triggers	Retention action triggered only after the customer signals an intent to leave.
Statistical	Traditional regression / chi-square analysis	Linear or logistic models on a small hand-picked set of variables, with strong distributional assumptions.
Statistical	Survival / hazard analysis	Time-to-event modelling of expected customer lifetime, usually at cohort rather than individual level.
Basic ML	Single decision trees / k-NN / Naive Bayes	Simple classifiers trained on raw features, usually without imbalance handling or tuning.
Basic ML	Accuracy-only classification studies	Academic projects that train a classifier and report accuracy without probability output, explanation or deployment.

9.2 Drawbacks of Existing Methods

Drawback	Explanation	Business Consequence
Reactive rather than predictive	Descriptive reporting and exit surveys detect churn only after the customer has already decided to leave.	Intervention arrives too late; the customer is effectively unrecoverable.
Poor handling of non-linearity	Rule-based and linear statistical methods cannot capture interaction effects among dozens of correlated attributes.	Genuine at-risk customers are missed while safe customers are wrongly flagged.
Low predictive accuracy and recall	Single classifiers without tuning or imbalance handling typically achieve high overall accuracy but very poor recall on the churn class.	The majority of actual churners go undetected, defeating the purpose of the system.
High false-positive rate	Coarse rules flag large numbers of loyal customers as at-risk.	Retention discounts are wasted on customers who would have stayed anyway, eroding margin.
Binary output with no prioritisation	A yes/no label carries no information about confidence or urgency.	The retention team cannot rank a limited-capacity call list; effort is spread thinly and randomly.
No value weighting	Existing systems treat every flagged customer as equally important.	High-value customers receive the same treatment as marginal ones; revenue

		protection is inefficient.
Black-box opacity	Where advanced models are used, they provide no per-customer justification.	Business users distrust and ultimately ignore the model; no targeted offer can be designed.
No prescriptive layer	Existing methods identify risk but do not recommend what to do about it.	Analytics output remains an interesting report rather than an operational workflow.
No financial quantification	Accuracy and ROC-AUC are not expressed in monetary terms.	Management cannot justify or size a retention budget; the project cannot demonstrate ROI.
Not deployed / not usable	Models remain in notebooks accessible only to the data scientist.	Zero operational adoption; the analytical investment produces no return.
Poor scalability and maintainability	Manual rules must be rewritten whenever the market, tariffs or customer behaviour change.	The system degrades over time and becomes a maintenance burden.

10. Research Gaps Identified

Consolidating the drawbacks of practice with the limitations observed in the reviewed literature, the following specific gaps were identified. Each gap is directly addressed by a corresponding component of the proposed system.

Gap #	Gap Identified	Evidence from Survey	How This Project Addresses It
G1	Insufficient like-for-like benchmarking of interpretable and ensemble models on a common pipeline	Studies typically favour one family of models; preprocessing differs between papers, making comparison unreliable	Logistic Regression, Random Forest, XGBoost and SVM are trained on an identical preprocessing pipeline and split, and compared on six metrics
G2	Class imbalance frequently under-addressed	Several works report accuracy without recall or PR-AUC on the minority churn class	Stratified splitting, class weighting and threshold tuning are applied; PR-AUC and recall are reported alongside accuracy
G3	Churn treated as a binary label rather than a probability	Most comparative studies output hard labels only	The system outputs a calibrated churn probability per customer and banded risk tiers for prioritisation
G4	Segmentation and prediction treated as separate exercises	Segmentation papers omit prediction; prediction papers omit segmentation	K-Means segmentation is integrated into the pipeline and every prediction is reported in its segment context
G5	Explainability limited to global feature importance	Many works report only aggregate importance rankings	Global importance is combined with per-customer SHAP reason codes surfaced directly in the dashboard
G6	No prescriptive bridge from explanation to action	Explainability studies end at attribution and stop short of intervention design	A rule-mapped recommendation engine converts each dominant risk driver into a specific retention offer or action
G7	Absence of financial impact quantification	Accuracy and AUC are reported; revenue and ROI almost never are	A business impact module computes revenue at risk, expected saved revenue, campaign ROI and projected churn-rate reduction
G8	Lack of value-weighted	Risk ranking ignores how much each	Risk score is combined with an

	prioritisation	customer is worth	ARPU-based customer value proxy to produce a value-weighted priority list
G9	Little or no deployment; no interactive analytics interface	The overwhelming majority of reviewed works remain offline experiments	The complete pipeline is deployed as an interactive Streamlit dashboard with upload, scoring, drill-down and export
G10	Weak feature engineering; raw columns used as-is	Few studies construct business-meaningful derived variables	Domain-driven features (tenure bands, ARPU, charge ratio, service count, contract risk flag) are engineered and evaluated

11. Proposed System

The proposed system is an integrated Customer Churn Prediction and Retention Analytics platform. It is deliberately designed as a business decision-support product rather than a standalone classifier: every analytical component terminates in an output that a retention manager can act upon. The platform is organised into eleven functional modules spanning four architectural layers.

11.1 Functional Modules

Module	Component	Function	Business Output
M1	Data Cleaning & Exploratory Analysis	Missing-value treatment, type correction, duplicate and outlier handling; univariate, bivariate and correlation analysis of churn behaviour	A verified analysis-ready dataset and a baseline churn profile of the customer base
M2	Customer Segmentation	K-Means clustering over RFM-style value indicators (tenure, service count, monthly charges) with elbow/silhouette-based selection of K	Named customer segments (e.g. 'High-Value Loyal', 'New & Fragile', 'Low-Value Price-Sensitive') with segment-wise churn rates
M3	Feature Engineering	Construction of tenure bands, ARPU, total-to-monthly charge ratio, subscribed-service count, contract risk flag and value tier; encoding and scaling	A richer, business-interpretable feature set that improves model lift
M4	Logistic Regression Model	Regularised logistic regression as an interpretable, well-calibrated benchmark	Transparent odds-ratio view of which attributes raise or lower churn odds
M5	Random Forest / XGBoost Models	Bagged and gradient-boosted tree ensembles with hyperparameter tuning and class weighting	High-accuracy churn detection capturing non-linear behavioural interactions
M6	Model Comparison Engine	Uniform evaluation of all models on accuracy, precision, recall, F1, ROC-AUC and PR-AUC with cross-validation	Evidence-based selection and justification of the production model
M7	Churn Probability Prediction	Calibrated probability output $p(\text{churn})$ for each customer, banded into Low (<0.35), Medium ($0.35\text{--}0.65$) and High (>0.65) risk tiers	A ranked, prioritised watch-list instead of an undifferentiated yes/no flag
M8	Explainable AI / Feature Importance	Global feature importance plus SHAP local attribution converted into plain-language reason codes	A defensible 'why' for every flagged customer, enabling a targeted rather than generic response
M9	High-Risk Customer Dashboard	Streamlit interface with KPI tiles, segment charts, model comparison plots, sortable high-risk table, drill-down explanation panel and CSV export	An operational daily work-list for the retention team
M10	Business Recommendation Engine	Rule-based mapping from dominant SHAP drivers and customer segment to a specific retention intervention	Concrete prescribed action per customer (contract upgrade offer, support intervention, loyalty reward, price re-plan)
M11	Revenue & Retention	Computation of revenue at risk, expected	A financial business case that

	Impact Model	saved revenue, campaign cost, net benefit, ROI and projected churn-rate reduction	management can approve and measure
--	--------------	---	------------------------------------

11.2 Novelty of the Proposed System

- **Unified pipeline** — Segmentation, prediction, explanation, recommendation and financial quantification operate on one consistent data pipeline rather than as disconnected experiments.
- **Probability-first design** — Every downstream business decision is driven by a calibrated probability, not a hard label.
- **Risk $\times$ Value prioritisation** — At-risk customers are ranked by expected revenue at risk, so retention effort is allocated where it protects the most money.
- **Explanation-driven prescription** — Retention offers are selected from the customer's own dominant churn drivers rather than applied uniformly.
- **Financially closed loop** — The system reports the rupee value of the churn it expects to prevent and the ROI of the campaign required to prevent it.

12. System Architecture

The system follows a four-layer architecture. The Data Layer ingests and stores customer records; the Data Preparation and Analytics Layer cleans, enriches and segments them; the Machine Learning and Explainability Layer trains, compares and interprets predictive models; and the Business Intelligence and Action Layer converts model output into risk scores, dashboards, recommendations and financial estimates that flow back to the business user.

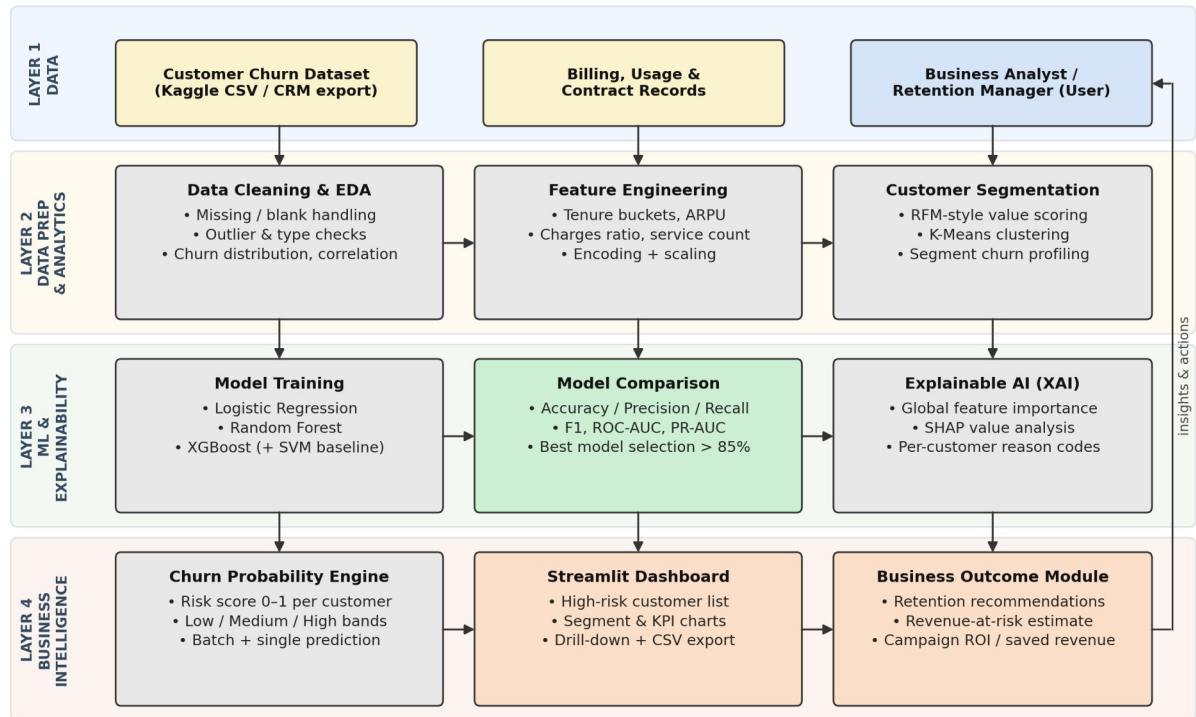


Figure 1: Four-layer system architecture of the churn prediction and retention analytics platform

12.1 Layer Responsibilities

Layer	Components	Responsibility
Layer 1 — Data	Kaggle churn dataset / CRM export, billing and contract records, business user	Ingest structured customer data and accept user-uploaded files through the application
Layer 2 — Data Preparation & Analytics	Data cleaning and EDA, feature engineering, K-Means customer segmentation	Produce an analysis-ready, feature-enriched and segmented customer table
Layer 3 — ML & Explainability	Model training (LR, RF, XGBoost, SVM), model comparison, SHAP-based XAI	Train and benchmark models, select the best performer and generate global and local explanations
Layer 4 — Business Intelligence & Action	Churn probability engine, Streamlit dashboard, business outcome module	Score, rank, visualise, recommend and quantify the financial impact of retention actions

13. Detailed Methodology

The project follows a comparative-analytics research methodology executed in nine sequential phases. The complete workflow is shown in Figure 2.

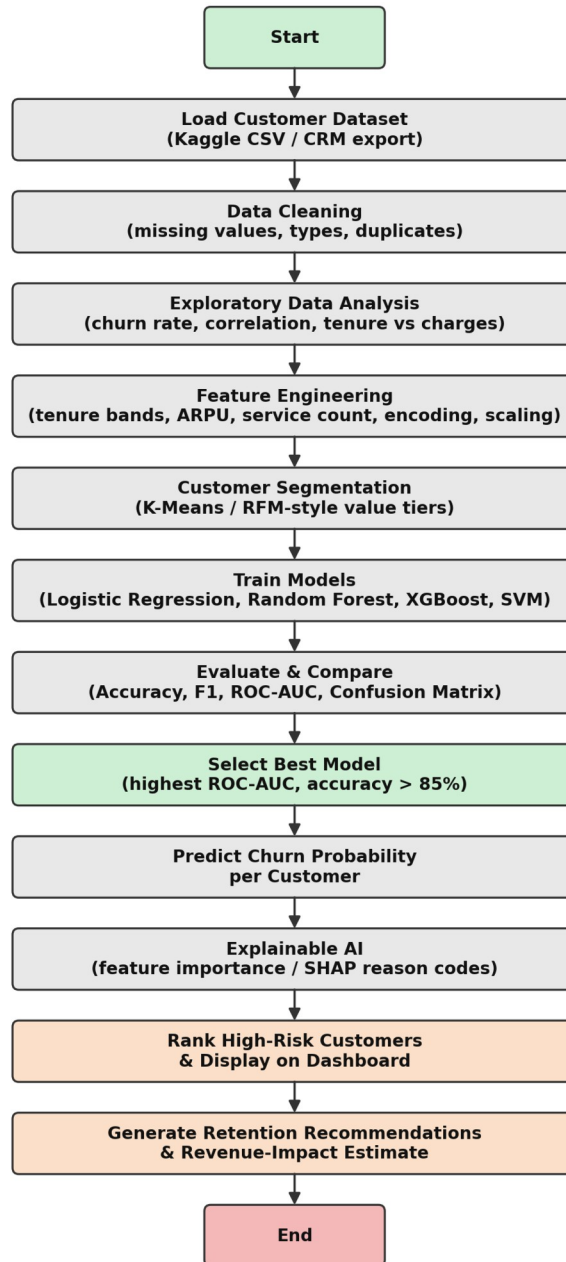


Figure 2: End-to-end project workflow from raw data to quantified business recommendation

13.1 Phase 1 — Data Collection and Understanding

- Acquire the public customer churn dataset (approximately 7,000 customer records, 20+ attributes) covering demographics, subscribed services, contract terms, billing information and the churn label.
- Profile the dataset: record count, attribute types, cardinality of categorical variables, and the baseline churn rate (approximately 26.5% in the reference dataset).
- Define the target variable (Churn: Yes = 1, No = 0) and separate identifier columns from predictors.

13.2 Phase 2 — Data Cleaning

- Replace blank strings with null values and convert incorrectly typed numeric columns (notably TotalCharges, which is stored as text) into numeric form.
- Impute or remove records with missing values; blank TotalCharges entries correspond to newly joined customers with zero tenure and are handled explicitly rather than dropped blindly.
- Remove duplicate customer records and validate value ranges for tenure and charges.
- Detect outliers using the IQR method and treat them by capping rather than deletion to preserve rare but genuine high-value customers.
- Standardise categorical labels (for example, unifying 'No internet service' and 'No' where semantically equivalent).

13.3 Phase 3 — Exploratory Data Analysis (EDA)

- **Univariate analysis** — Distribution of churn, tenure, monthly charges and contract types.
- **Bivariate analysis** — Churn rate by contract type, payment method, internet service, senior-citizen status and tech-support subscription.
- **Correlation analysis** — Heatmap of numerical attributes to detect multicollinearity (for example between tenure and total charges).
- **Behavioural analysis** — Tenure versus monthly charges scatter coloured by churn, revealing the high-charge / low-tenure danger zone.
- **Cohort analysis** — Churn rate by tenure band to quantify early-life churn risk.

13.4 Phase 4 — Feature Engineering

Engineered Feature	Definition	Business Rationale
Tenure Band	Tenure bucketed into 0–6, 7–12, 13–24, 25–48 and 49+ months	Churn risk is highly concentrated in the first year; banding captures this non-linearity explicitly
ARPU (Average Revenue Per User)	$\text{TotalCharges} \div \text{tenure}$ (guarded for zero tenure)	Normalised revenue contribution; a direct proxy for customer value
Charge Ratio	$\text{MonthlyCharges} \div (\text{TotalCharges} \div$	Detects recent price increases — a well-known

	tenure)	churn trigger
Service Count	Count of subscribed add-on services (security, backup, protection, support, streaming)	Higher service adoption increases switching cost and correlates with loyalty
Contract Risk Flag	Binary flag for month-to-month contracts	Commitment length is consistently the single strongest churn predictor in the literature
Value Tier	Quartile ranking of customers by ARPU	Enables value-weighted prioritisation of retention effort
Auto-Pay Flag	Binary flag for automatic payment methods	Manual payment methods (notably electronic cheque) are associated with materially higher churn

Categorical variables are encoded (one-hot encoding for nominal attributes, ordinal encoding where a natural order exists) and all numerical features are standardised using StandardScaler so that distance-based and regularised algorithms are not dominated by large-magnitude attributes.

13.5 Phase 5 — Customer Segmentation

- Construct RFM-style proxies from the available attributes: tenure as a recency/longevity proxy, service count as a frequency/engagement proxy, and monthly charges as the monetary proxy.
- Standardise these three dimensions and apply K-Means clustering.
- Select the optimal number of clusters K using the elbow method on within-cluster sum of squares, confirmed by silhouette score (expected K in the range 3–5).
- Profile each cluster by size, mean tenure, mean monthly charges, service adoption, revenue contribution and observed churn rate.
- Assign business-readable names to the resulting segments, for example: 'High-Value Loyal', 'High-Value At-Risk', 'New & Fragile', and 'Low-Value Price-Sensitive'.
- Append the segment label as an additional feature available to the predictive models and as a filter dimension in the dashboard.

13.6 Phase 6 — Model Development

Model	Configuration	Purpose in the Study
Logistic Regression	L2 regularisation, balanced class weights, liblinear/lbfgs solver	Interpretable, well-calibrated benchmark; supplies odds-ratio interpretation of each driver
Random Forest	200 estimators, max depth 10, balanced class weights, fixed random state	Robust bagged ensemble resistant to overfitting; provides impurity-based feature importance
XGBoost	200 estimators, learning rate 0.1, max depth 6, scale_pos_weight for imbalance, logloss evaluation metric	Gradient-boosted ensemble expected to deliver the highest discriminative performance

Support Vector Machine	RBF kernel, probability estimation enabled	Margin-based baseline for comparison against tree ensembles
------------------------	--	---

- Split the data using a stratified 80/20 train–test split to preserve the churn class ratio.
- Apply 5-fold stratified cross-validation on the training set for reliable performance estimation.
- Tune hyperparameters using grid or randomised search optimised for ROC-AUC rather than accuracy.
- Address class imbalance through class weighting and, where beneficial, SMOTE-family resampling applied inside the cross-validation folds to avoid data leakage.

13.7 Phase 7 — Model Evaluation and Comparison

Metric	Definition	Why It Matters for Churn
Accuracy	Proportion of all predictions that are correct	Headline measure, but misleading on its own under class imbalance
Precision	Of customers predicted to churn, the fraction that actually churn	Controls wasted retention spend on false alarms
Recall (Sensitivity)	Of customers who actually churn, the fraction correctly identified	The critical metric — a missed churner is lost revenue
F1-Score	Harmonic mean of precision and recall	Balances wasted budget against missed churners
ROC-AUC	Area under the receiver operating characteristic curve	Threshold-independent measure of ranking quality; the primary model-selection criterion
PR-AUC	Area under the precision–recall curve	More informative than ROC-AUC when the positive class is a minority
Confusion Matrix	Counts of true/false positives and negatives	Exposes the concrete operational trade-off at the chosen decision threshold

Models are compared on a single consolidated results table and on overlaid ROC and precision–recall curves. The model with the highest ROC-AUC, subject to an accuracy above 85% and acceptable recall on the churn class, is promoted to production. The decision threshold is then tuned explicitly — rather than left at the default 0.5 — to reflect the relative business cost of a missed churner versus a wasted offer.

13.8 Phase 8 — Churn Probability Prediction and Explainable AI

- Generate a calibrated churn probability $p(\text{churn})$ for every customer using `predict_proba`, with probability calibration (Platt scaling or isotonic regression) verified through a calibration curve.
- Band customers into risk tiers: Low ($p < 0.35$), Medium ($0.35 \leq p \leq 0.65$) and High ($p > 0.65$).
- Compute global feature importance from the selected model to establish the overall churn drivers across the customer base.

- Apply SHAP (SHapley Additive exPlanations) to obtain local, per-customer attribution values showing how much each attribute pushed that specific customer's probability up or down.
- Translate the top contributing SHAP features into plain-language reason codes such as 'Month-to-month contract (+18% risk)', 'Tenure below 6 months (+12% risk)' or 'No technical support (+7% risk)'.
- Validate that the explanations are consistent with the EDA findings and with the Logistic Regression odds ratios, providing a cross-check on model trustworthiness.

13.9 Phase 9 — Business Recommendation and Revenue Impact

The final phase converts analytical output into a financial business case. The logic is shown in Figure 3.

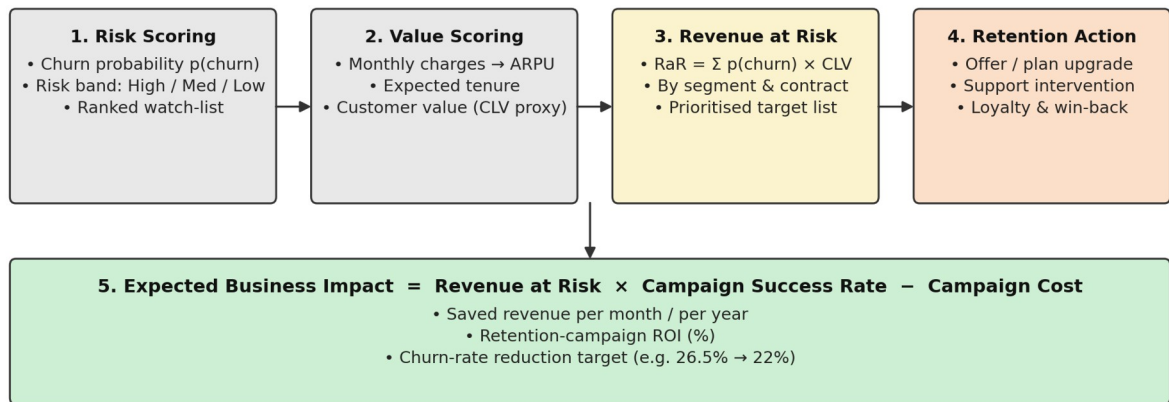


Figure 3: Business value chain converting churn probability into quantified revenue impact

13.9.1 Recommendation Mapping

Dominant Risk Driver (from SHAP)	Recommended Retention Action	Target Segment
Month-to-month contract	Offer a discounted 12- or 24-month contract with a loyalty bonus	All risk tiers; prioritise high-value customers
Tenure below 6 months	Structured onboarding programme, welcome call and first-90-days offer	New & Fragile segment
High monthly charges relative to segment	Plan right-sizing or a personalised tariff review	Low-Value Price-Sensitive segment
No technical support / online security	Complimentary trial of support and security add-ons	High-Value At-Risk segment
Electronic cheque / manual payment	Incentivised migration to auto-pay with a billing discount	All segments
Fibre-optic service with high charges	Service-quality audit followed by targeted compensation or a loyalty credit	High-Value At-Risk segment

Low service adoption	Bundled cross-sell offer to increase engagement and switching cost	Low-engagement customers across segments
----------------------	--	--

13.9.2 Revenue and Retention Impact Model

For each customer i , the following quantities are computed:

Customer Value (CLV proxy) $_i$ = Monthly Charges $_i$ × Expected Remaining Tenure (months)

Revenue at Risk $_i$ = $p(\text{churn})_i$ × Customer Value $_i$

Total Revenue at Risk = Σ_i Revenue at Risk $_i$ (over all high-risk customers)

Expected Saved Revenue = Total Revenue at Risk × Campaign Success Rate

Net Benefit = Expected Saved Revenue – (Number Targeted × Cost per Retention Offer)

Retention ROI (%) = (Net Benefit ÷ Campaign Cost) × 100

Projected Churn Rate = Baseline Churn Rate × (1 – Recall × Campaign Success Rate)

These quantities are recomputed live in the dashboard as the user adjusts the risk threshold, the assumed campaign success rate and the cost per offer, allowing the retention manager to explore the trade-off between reach and profitability before committing budget.

13.9.3 Illustrative Impact Calculation

Using the reference dataset as an example (7,043 customers, 26.5% baseline churn, average monthly charge of approximately US\$64.76), and assuming a model recall of 0.80 on the churn class and a retention campaign that succeeds with 30% of those contacted at a cost of US\$15 per offer:

Quantity	Calculation	Illustrative Value
Customers in the base	Given	7,043
Baseline churners per cycle	$7,043 \times 0.265$	$\approx 1,866$
Churners correctly identified	$1,866 \times 0.80$ recall	$\approx 1,493$
Average customer value (12-month horizon)	$\text{US\$}64.76 \times 12$	$\approx \text{US\$}777$
Revenue at risk among identified churners	$1,493 \times \text{US\$}777$	$\approx \text{US\$}1.16$ million
Customers actually retained	$1,493 \times 0.30$ success rate	≈ 448
Gross revenue saved	$448 \times \text{US\$}777$	$\approx \text{US\$}348,000$
Campaign cost	$1,493 \times \text{US\$}15$	$\approx \text{US\$}22,400$
Net benefit	$\text{US\$}348,000 - \text{US\$}22,400$	$\approx \text{US\$}325,600$
Retention ROI	$(325,600 \div 22,400) \times 100$	$\approx 1,453\%$
Projected churn rate	$26.5\% \times (1 - 0.80 \times 0.30)$	$\approx 20.1\%$

--	--	--

Note: the figures above are illustrative and are computed from stated assumptions; actual values will be produced by the deployed system from the live dataset and from campaign parameters supplied by the business.

13.10 Phase 10 — Deployment

- Serialise the selected model, the fitted scaler and the encoders as .pkl artefacts so that inference reproduces the exact training-time transformations.
- Build the Streamlit application with distinct pages: Overview KPIs, Exploratory Analysis, Segmentation, Model Comparison, High-Risk Customers, Explainability, and Business Impact.
- Support two prediction modes: batch scoring of an uploaded customer file, and single-customer what-if prediction through an input form.
- Provide CSV export of the prioritised high-risk target list, including risk band, segment, reason codes and recommended action for each customer.
- Validate the deployed application on a held-out sample and document the run instructions in the project README.

14. Dataset Description

The primary dataset used in this study is the publicly available Customer Churn Prediction dataset hosted on Kaggle. It is a cross-sectional, structured dataset of service subscribers containing demographic, account, service and billing attributes together with a binary churn label, and it is widely used as a benchmark in the churn-prediction literature.

Dataset source: <https://www.kaggle.com/datasets/deaneshtakkallapati/customer-churn-prediction/data>

Attribute Group	Representative Fields	Type
Customer identity	customerID	Identifier (excluded from modelling)
Demographics	gender, SeniorCitizen, Partner, Dependents	Categorical / binary
Account information	tenure, Contract, PaperlessBilling, PaymentMethod	Numerical & categorical
Services subscribed	PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies	Categorical
Billing	MonthlyCharges, TotalCharges	Numerical (TotalCharges requires type correction)
Target	Churn (Yes / No)	Binary categorical → encoded to 1 / 0

14.1 Known Data Characteristics

- Approximately 7,000 customer records with more than 20 attributes.
- Baseline churn rate of roughly 26.5%, giving a moderately imbalanced classification problem.
- TotalCharges is stored as text and contains blank entries for customers with zero tenure, requiring explicit cleaning.
- Tenure and TotalCharges are strongly correlated, so multicollinearity must be considered for the Logistic Regression model.
- The dataset is anonymised and contains no personally identifiable information, so no additional privacy treatment is required for academic use.

15. Hardware and Software Requirements

15.1 Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i5 (8th generation) or AMD Ryzen 5	Intel Core i7 / Ryzen 7 or higher
RAM	8 GB	16 GB

Storage	10 GB free disk space	20 GB free SSD space
Display	1366 × 768	1920 × 1080 or higher (for dashboard readability)
GPU	Not required (classical ML on tabular data)	Optional; useful only for future deep-learning extensions
Network	Stable internet connection for package installation and dataset download	Broadband connection; required if the dashboard is hosted remotely

15.2 Software Requirements

Category	Software / Library	Version	Purpose
Operating System	Windows 10/11, Ubuntu 20.04+ or macOS	—	Development and deployment platform
Language	Python	3.8 or higher	Core implementation language
IDE / Environment	Jupyter Notebook, VS Code, Google Colab	Latest	Development, experimentation and documentation
Data handling	NumPy, Pandas	Latest stable	Numerical computation and data manipulation
Machine learning	Scikit-learn	1.0+	Preprocessing, Logistic Regression, Random Forest, SVM, K-Means, metrics
Gradient boosting	XGBoost	1.6+	Gradient-boosted tree classifier
Imbalance handling	Imbalanced-learn (imblearn)	Latest stable	SMOTE / SMOTE-ENN resampling for the minority churn class
Explainable AI	SHAP	0.41+	Global and local model explanation and reason-code generation
Visualisation	Matplotlib, Seaborn, Plotly	Latest stable	EDA charts, evaluation curves and interactive dashboard visuals
Web application	Streamlit	1.20+	Interactive analytics dashboard and deployment layer
Model persistence	Pickle / Joblib	Standard library / latest	Serialisation of the model, scaler and encoders
Version control	Git / GitHub	Latest	Source-code management and project submission

15.3 Functional and Non-Functional Requirements

Type	Requirement
Functional	The system shall accept a customer dataset in CSV format through the dashboard.
Functional	The system shall clean, encode and scale the uploaded data using the stored training-time transformations.
Functional	The system shall assign each customer to a behavioural segment.
Functional	The system shall output a churn probability and a risk band for every customer.
Functional	The system shall display the top N high-risk customers ranked by revenue at risk.
Functional	The system shall display per-customer reason codes explaining each prediction.

Functional	The system shall recommend a retention action for each high-risk customer.
Functional	The system shall compute revenue at risk, expected saved revenue and campaign ROI.
Functional	The system shall allow export of the prioritised target list as a CSV file.
Non-Functional	Model accuracy shall exceed 85% with ROC-AUC above 0.85 on the held-out test set.
Non-Functional	Batch scoring of 10,000 customer records shall complete within 30 seconds on the recommended hardware.
Non-Functional	The dashboard shall be usable by a business analyst with no programming knowledge.
Non-Functional	Every prediction shall be accompanied by an explanation (no unexplained black-box output).
Non-Functional	The pipeline shall be reproducible through fixed random seeds and serialised artefacts.

16. Applications

Sector	Application of the System	Business Benefit
Telecommunications	Predicting subscriber churn across prepaid and postpaid plans; identifying at-risk high-ARPU subscribers before contract renewal	Protects recurring subscription revenue in a saturated, price-competitive market
Banking & Financial Services	Detecting attrition among savings, credit-card and loan customers; flagging accounts trending towards dormancy	Preserves deposit balances, fee income and cross-sell opportunity
Insurance	Predicting non-renewal of motor, health and home policies ahead of the renewal window	Increases renewal rates and stabilises the underwriting book
SaaS & Subscription Software	Identifying accounts with declining engagement and predicting non-renewal at seat or account level	Improves net revenue retention, a primary SaaS valuation driver
OTT / Streaming Media	Forecasting subscription cancellations after content cycles or price changes	Enables targeted content and pricing interventions before cancellation
E-Commerce & Retail	Detecting lapsing customers in loyalty programmes using recency and frequency behaviour	Supports reactivation campaigns and protects repeat-purchase revenue
Utilities & Energy Retail	Predicting switching in deregulated electricity and gas markets	Reduces customer switching losses and acquisition re-spend
Healthcare & Fitness	Predicting membership lapse in gyms, clinics and wellness subscriptions	Improves membership continuity and facility utilisation
EdTech	Predicting course drop-off and non-renewal of learning subscriptions	Enables timely learner intervention and improves completion rates
B2B Services	Predicting contract non-renewal for managed-service and maintenance agreements	Protects high-value annual contracts through early account management

17. Expected Output

17.1 Analytical Outputs

- A cleaned, feature-engineered and segmented customer dataset ready for analysis.
- A complete EDA report covering churn distribution, correlation structure, churn rate by contract, payment method and tenure band, and the tenure-versus-charges risk map.
- Customer segment profiles (expected 3–5 segments) with size, average tenure, average monthly charges, revenue contribution and churn rate per segment.
- A consolidated model comparison table reporting accuracy, precision, recall, F1-score, ROC-AUC and PR-AUC for Logistic Regression, Random Forest, XGBoost and SVM.
- Confusion matrices, overlaid ROC curves and precision–recall curves for all four models.
- A documented best-model selection with justification, targeting accuracy above 85% and ROC-AUC above 0.85 (XGBoost or Random Forest expected to lead based on the literature).

- A global feature-importance ranking and SHAP summary plot identifying the dominant churn drivers, expected to include contract type, tenure, monthly charges, payment method and tech-support status.

17.2 Predictive Outputs

- A calibrated churn probability between 0 and 1 for every customer in the dataset.
- Risk banding of the entire customer base into Low, Medium and High risk tiers.
- A ranked high-risk customer table sorted by revenue at risk rather than by probability alone.
- Per-customer SHAP reason codes stating the top three factors driving that customer's risk.
- Single-customer what-if prediction allowing a user to alter an attribute (for example, contract type) and observe the resulting change in churn probability.

17.3 Business Outputs

- An interactive Streamlit dashboard with KPI tiles (total customers, baseline churn rate, high-risk count, total revenue at risk), segment charts, model comparison plots and a drill-down explanation panel.
- A prescribed retention action for each high-risk customer, derived from that customer's own dominant risk drivers and segment.
- A quantified financial summary: total revenue at risk, expected saved revenue, campaign cost, net benefit and retention ROI.
- A projected reduction in the overall churn rate (illustratively from approximately 26.5% to approximately 20–22% under the stated assumptions).
- An exportable CSV target list ready to be handed to the retention or customer-success team.

17.4 Project Deliverables

- Modular, documented Python source code (data loading, preprocessing, EDA, segmentation, feature engineering, model training, evaluation, explainability, prediction and business-impact modules).
- Serialised model artefacts: `best_model.pkl`, `scaler.pkl` and `label_encoder.pkl`.
- The deployed Streamlit application together with run instructions.
- This project report, a presentation deck and a README file.

18. Advantages

Advantage	Description
High predictive accuracy	Benchmarked ensemble models target accuracy above 85% and ROC-AUC above 0.85, substantially outperforming rule-based and single-classifier approaches.
Proactive rather than reactive	Customers are identified before they leave, converting churn management from post-mortem reporting into preventive action.
Probability-based	Continuous risk scores allow a limited retention team to work the highest-risk

prioritisation	customers first instead of an undifferentiated flagged list.
Value-weighted targeting	Combining churn probability with customer value directs retention budget to the customers whose loss would cost the most.
Full transparency	SHAP-based reason codes mean no prediction is a black box; every flag can be justified to a manager, an auditor or the customer-facing agent.
Prescriptive, not merely predictive	The system recommends a specific retention action per customer instead of leaving interpretation to the business.
Financially quantified	Revenue at risk, saved revenue and ROI are reported in currency terms, allowing management to size and justify the retention budget.
Segment-aware strategy	Clustering enables differentiated treatment of high-value, new and price-sensitive customers rather than a single blanket offer.
Business-user accessible	The Streamlit dashboard requires no coding knowledge, ensuring operational adoption by analysts and retention managers.
Reusable and domain-portable	The pipeline is dataset-agnostic and transfers to banking, insurance, SaaS, retail and utilities with minimal reconfiguration.
Modular and maintainable	Clean separation of data, modelling, explanation and business layers makes the system easy to extend, retrain and audit.
Automated and scalable	Batch scoring of large customer files runs in seconds and can be scheduled, removing manual analysis effort.
Cost-effective	Built entirely on open-source Python libraries with no licensing cost and no GPU requirement.
Academically rigorous	Stratified cross-validation, explicit imbalance handling, multi-metric evaluation and threshold tuning ensure the reported results are defensible.

19. Work Breakdown Structure and Timeline

Phase	Work Package	Key Activities	Deliverable	Duration
1	Problem definition & literature survey	Define scope and objectives; survey 2022–2026 literature; identify gaps	Problem statement, literature survey table, gap analysis	1 week
2	Dataset acquisition & understanding	Download dataset; profile attributes; establish baseline churn rate	Data profile report	0.5 week
3	Data cleaning & EDA	Handle missing values and types; outlier treatment; univariate, bivariate and correlation analysis	Cleaned dataset and EDA notebook with visualisations	1 week
4	Feature engineering	Construct tenure bands, ARPU, charge ratio, service count, contract-risk and value-tier features; encode and scale	Feature-engineered dataset and feature dictionary	1 week
5	Customer segmentation	Build RFM proxies; run K-Means; select K via elbow and silhouette; profile and name segments	Segment profiles and segment-wise churn analysis	1 week
6	Model development	Train Logistic Regression, Random Forest, XGBoost and SVM; tune hyperparameters; handle imbalance	Trained model set with tuning logs	1.5 weeks
7	Model evaluation & selection	Cross-validate; compute six metrics; plot ROC and PR curves; tune the decision threshold; select the best model	Model comparison report and serialised best model	1 week
8	Explainable AI	Compute global feature importance; generate SHAP values; build the reason-code generator	SHAP plots and per-customer reason-code module	1 week
9	Business logic layer	Implement risk banding, recommendation mapping, revenue-at-risk and ROI calculations	Business impact module	1 week
10	Dashboard development	Build the multi-page Streamlit application; integrate scoring, explanation, recommendation and export	Deployed interactive dashboard	1.5 weeks
11	Testing & validation	Validate the pipeline end to end; test batch and single-customer prediction; verify reproducibility	Test report	0.5 week
12	Documentation & reporting	Compile the final report, presentation and README	Project report, slide deck and README	1 week

Total estimated duration: approximately 12 weeks.

20. Conclusion

This project presents a complete, business-oriented artificial intelligence and analytics solution for customer churn prediction and retention management in service-oriented organisations. Rather than treating churn prediction as an isolated classification exercise, the work deliberately spans the entire analytical value chain: from raw customer data, through cleaning, exploratory analysis, business-driven feature engineering and unsupervised customer segmentation, to supervised model benchmarking, calibrated probability estimation, explainable attribution, prescriptive recommendation and quantified financial impact.

Four supervised classifiers — Logistic Regression, Random Forest, XGBoost and a Support Vector Machine baseline — are trained on an identical pipeline and compared across six evaluation metrics, ensuring that model selection is evidence-based rather than assumed. The literature surveyed indicates that gradient-boosted ensembles should lead this comparison, while the inclusion of Logistic Regression guarantees an interpretable and well-calibrated benchmark. Class imbalance, which is intrinsic to churn data and frequently neglected in comparable studies, is handled explicitly through stratification, class weighting and threshold tuning, with recall and PR-AUC reported alongside accuracy so that the model is judged on its ability to find churners rather than on its ability to identify the loyal majority.

The distinguishing contribution of the work lies beyond the classifier. By producing a calibrated churn probability for every individual customer, the system enables ranked, capacity-aware prioritisation. By applying SHAP-based explainability, it attaches a defensible reason to every flagged customer, transforming an opaque score into an actionable diagnosis. By combining risk with customer value, it directs finite retention budget towards the customers whose departure would cost the most. By mapping each dominant risk driver to a specific retention intervention, it converts prediction into prescription. And by computing revenue at risk, expected saved revenue and retention-campaign ROI, it expresses the entire analysis in the language that management uses to make decisions.

Deployment through an interactive Streamlit dashboard completes the transition from experiment to data product. A retention manager with no programming background can upload a customer file, view segment-level churn behaviour, examine the ranked high-risk list with its reason codes and recommended actions, adjust campaign assumptions and export a prioritised target list — all without touching code.

In summary, the proposed system addresses the ten research and practice gaps identified in this report and delivers a scalable, transparent, reusable and financially grounded churn management framework. It demonstrates that the value of applied machine learning in business is realised not at the point where a model reports its accuracy, but at the point where its output changes a decision. Future work will extend the system towards live CRM integration, automated retraining with drift monitoring, full customer lifetime value modelling, and uplift modelling to isolate genuinely persuadable customers — further sharpening the return on every rupee of retention spend.

References

- [1] Habib, M. U., Aslam, S., Ghafoor, I., & Shahzad, M. K. (2025). Enhancing telecom customer churn prediction using hybrid ensemble learning and class imbalance handling. SSRN Electronic Journal. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5956757
- [2] Enhancing customer churn prediction in telecommunications: An adaptive ensemble learning approach (2024). arXiv preprint arXiv:2408.16284. <https://arxiv.org/abs/2408.16284>
- [3] Explaining customer churn prediction in the telecom industry using tabular machine learning models (2024). Machine Learning with Applications, Elsevier. <https://www.sciencedirect.com/science/article/pii/S2666827024000434>
- [4] A comprehensive evaluation of machine learning and deep learning models for churn prediction (2025). Information (MDPI), 16(7), 537. <https://www.mdpi.com/2078-2489/16/7/537>
- [5] Prediction of bank credit customers' churn based on machine learning and interpretability analysis (2025). Data Science in Finance and Economics (AIMS Press), 5(1). <https://www.aimspress.com/article/id/678b834bba35de451fe3ccc5>
- [6] Explainable AI-driven customer churn prediction: A multi-model ensemble approach with SHAP-based feature analysis (2026). Scientific Reports. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12929532/>
- [7] Customer churn prediction for telecommunication companies using machine learning and ensemble methods (2024). Engineering, Technology & Applied Science Research (ETASR). <https://etasr.com/index.php/ETASR/article/view/7480>
- [8] Enhancing customer retention in online retail through churn prediction: A hybrid RFM, K-means, and deep neural network approach (2025). Expert Systems with Applications, Elsevier. <https://www.sciencedirect.com/science/article/abs/pii/S0957417425020846>
- [9] Customer segmentation: Automatic K-optimization and RFM-based K-means clustering (2025). Proceedings of the 10th International Conference on Intelligent Information Technology, ACM. <https://dl.acm.org/doi/full/10.1145/3731763.3731805>
- [10] A novel hybrid deep learning framework for customer churn prediction using RFM and embedding clustering (2026). Scientific Reports. <https://www.nature.com/articles/s41598-026-53220-0>
- [11] Customer segmentation using an extended RFM model and clustering algorithms in e-commerce (2026). Journal of Theoretical and Applied Electronic Commerce Research (MDPI), 21(5), 142. <https://www.mdpi.com/0718-1876/21/5/142>
- [12] Explainable customer churn prediction in telecom using machine learning (2025). Ingénierie des Systèmes d'Information, IIETA. <https://www.iieta.org/download/file/fid/191322>
- [13] An explainable AI approach for telecom churn prediction and retention strategy design using XGBoost and SHAP (2025). ResearchGate. <https://www.researchgate.net/publication/395617319>
- [14] Wei, Y. (2025). Comparative analysis of machine learning models for telecom customer churn prediction. Theoretical and Natural Science, 134, 24–30.
- [15] Integration of churn rate prediction and customer lifetime value estimation using temporal convolution networks (2024). International Journal of Engineering Research & Technology (IJERT).

<https://www.ijert.org/integration-of-churn-rate-prediction-and-customer-lifetime-value-estimation-using-temporal-convolution-networks>

- [16] Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems (NeurIPS)*, 30, 4765–4774.
- [17] Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794.
- [18] Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.
- [19] Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16, 321–357.
- [20] Takkallapati, D. Customer Churn Prediction dataset. Kaggle. <https://www.kaggle.com/datasets/deaneshtakkallapati/customer-churn-prediction/data>